Computational Physics Lectures

Week 02: Metropolis Monte Carlo

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Week 2

Metropolis Monte Carlo

2.1 Introduction

In statistical mechanics, an ensemble is an idealized collection of a large number of copies of a system, all prepared under the same macroscopic conditions but potentially occupying different microscopic states. Consider a system in thermal equilibrium with a heat bath at temperature T. The corresponding statistical description is given by the canonical ensemble. A central result of statistical physics is that the probability of finding the system in a microstate i with energy E_i is

$$P_i = \frac{1}{Z} \exp\left(-\frac{E_i}{k_B T}\right), \tag{2.1}$$

where k_B is the Boltzmann constant. The quantity Z is the partition function, which ensures normalization of the probability distribution:

$$Z = \sum_i \exp\left(-\frac{E_i}{k_B T}\right). \tag{2.2}$$

Using the probability distribution introduced above, the thermal average of an observable O is

$$\langle O \rangle = \sum_i O_i P_i = \frac{1}{Z} \sum_i O_i \exp\left(-\frac{E_i}{k_B T}\right). \tag{2.3}$$

2.2 Contributions to Z

The kinetic and potential contributions to the partition function can be separated when the Hamiltonian of the system can be expressed as a sum of a kinetic energy term that depends only on the momenta and a potential energy term that depends only on the positions. This situation often arises in classical mechanics as well as in quantum mechanics, but there may be exceptions, e.g. for charged particles moving in a magnetic field or when non-Cartesian coordinates are used.

For a classical system with N particles, the Hamiltonian H can often be written as:

$$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q}) \tag{2.4}$$

Here, $\mathbf{q}$ denotes the collection of generalized coordinates, and $\mathbf{p}$ denotes the corresponding conjugate momenta. For N particles in d spatial dimensions, each consists of dN components. $T(\mathbf{p})$ is the kinetic energy, which depends only on the momenta $\mathbf{p}$, and $V(\mathbf{q})$ is the potential energy, which depends only on the positions $\mathbf{q}$.

In this case, the classical partition function Z can be written as an integral over phase space (p and q) for N (distinguishable) particles and d dimensions:

$$Z = \frac{1}{h^{dN}} \int \exp\left(-\left[T(\mathbf{p}) + V(\mathbf{q})\right]/(k_B T)\right)\, d^{dN}p\, d^{dN}q, \tag{2.5}$$

where the units of Planck's constant h ensures that Z becomes dimensionless, while its value can be argued when comparing to quantum mechanics where phase space is discretized.

Because T depends only on $\mathbf{p}$ and V depends only on $\mathbf{q}$, the Boltzmann factor and the integral factorize:

$$Z = \frac{1}{h^{dN}} \left(\int \exp\left(-T(\mathbf{p})/(k_B T)\right)\, d^{dN}p \right) \left(\int \exp\left(-V(\mathbf{q})/(k_B T)\right)\, d^{dN}q \right). \tag{2.6}$$

Thus, the partition function factorizes into a kinetic part and a potential part:

$$Z = Z_{\text{kin}} \cdot Z_{\text{pot}}, \tag{2.7}$$

where we let the kinetic part contain the normalizing constant.

In the following we shall call the latter Q:

$$Q = Z_{\text{pot}}, \tag{2.8}$$

and we shall be more or less neglecting the kinetic contribution, which, however, for an ideal gas, or a set of decoupled oscillators, can be evaluated exactly to give:

$$Z_{\text{kin}} = \left(\frac{\sqrt{2\pi m k_B T}}{h} \right)^{dN}, \tag{2.9}$$

where d is the dimensionality and N is the number of particles.

2.3 Harmonic oscillator

The harmonic oscillator is a fundamental model in both classical and quantum mechanics, describing a system where a particle experiences a restoring force proportional to its displacement from an equilibrium position. The potential energy $V(x)$ of a one-dimensional harmonic oscillator is given by:

$$V(x) = \frac{1}{2} k x^2 \tag{2.10}$$

where k is the spring constant and x is the displacement from the equilibrium position.

For a classical harmonic oscillator in thermal equilibrium at temperature T, the probability distribution $P(x)$ of finding the particle at position x is given by the Boltzmann distribution:

$$P(x) = \frac{1}{Q} \exp\left(-\frac{V(x)}{k_B T}\right) \tag{2.11}$$

where Q is the partition function (pertaining to the positional degree of freedom), which normalizes the probability distribution.

Substituting the expression for the potential energy:

$$P(x) = \frac{1}{Q} \exp\left(-\frac{\frac{1}{2}kx^2}{k_B T}\right). \tag{2.12}$$

The partition function Q for the position variable x is given by:

$$Q = \int_{-\infty}^{\infty} \exp\left(-\frac{\frac{1}{2}kx^2}{k_B T}\right) dx. \tag{2.13}$$

This is a Gaussian integral, which evaluates to:

$$Q = \sqrt{\frac{2\pi k_B T}{k}}. \tag{2.14}$$

Therefore, the probability distribution $P(x)$ becomes:

$$P(x) = \sqrt{\frac{k}{2\pi k_B T}} \exp\left(-\frac{\frac{1}{2}kx^2}{k_B T}\right). \tag{2.15}$$

2.4 Direct integration

The thermal average of the potential energy of the harmonic oscillator follows from the continuous version of the discrete formula, Eq. (2.3), given above:

$$\langle V(x) \rangle = \int_{-\infty}^{\infty} V(x) P(x) dx. \tag{2.16}$$

Substituting in the expression for the potential and for the probability this expression evaluates to:

$$\langle V(x) \rangle = \frac{1}{2} k_B T. \tag{2.17}$$

Considering now the heat capacity:

$$C_{V,\text{pot}} = \frac{d\langle V(x) \rangle}{dT} = \frac{1}{2} k_B. \tag{2.18}$$

We thus recover the well-known result from statistical physics that each uncoupled quadratic term in the Hamiltonian contributes $\frac{1}{2}k_B$ to the heat capacity.

Note that the kinetic energy also contributes an uncoupled quadratic term to the Hamiltonian, leading to an additional contribution

$$C_{V,\text{kin}} = \frac{1}{2}k_B. \tag{2.19}$$

Hence, the total heat capacity of a classical one-dimensional harmonic oscillator is $C_V = k_B$.

2.5 Heat capacity from fluctuations

You may recall from your statistical physics course that the heat capacity "controls" the energy fluctuations of a system in thermal equilibrium. This offers another way to evaluate the heat capacity, namely:

$$C_{V,\text{pot}} = \frac{\langle V(x)^2\rangle - \langle V(x)\rangle^2}{k_B T^2}. \tag{2.20}$$

The obvious advantage of this formula is that the system need only be modelled at one temperature, as no temperature dependence needs to be established as with the first formula, Eq. (2.18).

A derivation of Eq. (2.20) is provided below. Readers who are primarily interested in the computational method may proceed directly to the next section.

To improve the readability of the proof, we start by introducing the inverse thermal energy:

$$\beta = \frac{1}{k_B T} \tag{2.21}$$

which has units of inverse energy.

We immediately have:

$$\frac{d\beta}{dT} = -\frac{1}{(k_B T)^2}\cdot k_B = -k_B\beta^2$$

which can be used when applying the chain rule in the original expression for the heat capacity:

$$C_{V,\text{pot}} = \frac{d\langle V(x)\rangle}{dT} = \frac{d\beta}{dT}\cdot\frac{d\langle V(x)\rangle}{d\beta} = -k_B\beta^2\frac{d\langle V(x)\rangle}{d\beta} \tag{2.22}$$

The last derivative in this expression can now be evaluated, while using:

$$\begin{aligned}\langle V(x)\rangle &= \frac{1}{Q}\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx \\ Q &= \int_{-\infty}^{\infty}\exp\left(-\beta V(x)\right)dx.\end{aligned}$$

We have:

$$\begin{aligned}\frac{d\langle V(x)\rangle}{d\beta} &= \frac{d}{d\beta}\left(\frac{1}{Q}\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx\right) \\ &= \frac{1}{Q}\frac{d}{d\beta}\left(\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx\right) - \frac{\langle V(x)\rangle}{Q}\frac{dQ}{d\beta}.\end{aligned} \tag{2.23}$$

The two derivatives are evaluated as follows:

$$\begin{aligned}\frac{d}{d\beta}\left(\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx\right) &= \int_{-\infty}^{\infty} V(x)\left(-V(x)\right)\exp\left(-\beta V(x)\right)dx \\ &= -\int_{-\infty}^{\infty} V(x)^2\exp\left(-\beta V(x)\right)dx\end{aligned}$$

and

$$\frac{dQ}{d\beta} = \frac{d}{d\beta}\left(\int_{-\infty}^{\infty}\exp\left(-\beta V(x)\right)dx\right) = -\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx$$

Inserting these two derivatives into the result above we get:

$$\begin{aligned}\frac{d\langle V(x)\rangle}{d\beta} &= \frac{1}{Q}\left(-\int_{-\infty}^{\infty} V(x)^2\exp\left(-\beta V(x)\right)dx\right) \\ &\quad -\frac{\langle V(x)\rangle}{Q}\left(-\int_{-\infty}^{\infty} V(x)\exp\left(-\beta V(x)\right)dx\right) \\ &= -\langle V(x)^2\rangle + \langle V(x)\rangle^2\end{aligned}$$

Finally, when substituting into the original expression we have:

$$\begin{aligned}\frac{d\langle V(x)\rangle}{dT} &= -k_B\beta^2\frac{d\langle V(x)\rangle}{d\beta} \\ &= k_B\beta^2\left[\langle V(x)^2\rangle - \langle V(x)\rangle^2\right] \\ &= \frac{1}{k_BT^2}\left[\langle V(x)^2\rangle - \langle V(x)\rangle^2\right].\end{aligned}$$

2.6 Monte Carlo sampling

For systems more complicated than a single harmonic oscillator, it may not be possible to evaluate Q, and hence the normalized distribution $P(x)$, analytically. In such cases, it is useful to generate a *sample* of positions, $\{x_i\}$. To ensure that the sample accurately represents the thermal distribution, the positions $\{x_i\}$ must be drawn according to the probability distribution $P(x)$. At first, this may appear to require prior knowledge of the normalized distribution $P(x)$. As we shall see, however, the sample can be generated using only an unnormalized probability density, without evaluating the partition function.

Once you have established the sample, $\{x_i\}$, you may estimate the value of the integral, Eq. (2.16), for the thermal average given above:

$$\begin{aligned}\langle V(x)\rangle &= \int_{-\infty}^{\infty} V(x)P(x)dx \\ &\simeq \frac{1}{N}\sum_{i=1}^{N} V(x_i).\end{aligned}$$

2.7 Metropolis Monte Carlo sampling

Metropolis Monte Carlo sampling offers a means to obtain a thermal sample knowing only that the probability distribution is proportional to the Boltzmann factor, $\exp(-V(x)/k_BT)$, but not knowing the normalization term, the partition function Q. In this method, you start with an initial position x_0 and propose a new position x' based on a symmetric proposal distribution $T(x'|x)$, which only depends on the current position x. The new position x' is accepted with a probability given by the ratio of the target distributions:

$$\text{Acceptance probability from } x \text{ to } x' = \min(1, \frac{P(x')}{P(x)}) \tag{2.24}$$

If the proposed position x' is accepted, it becomes the new current position; otherwise, the current position is retained. This process is repeated to generate a sequence of positions $\{x_i\}$ that sample the target distribution $P(x)$.

By *symmetric* proposal distribution is meant that the likelihood of proposing the move from x to x' is the same as that of proposing the move from x' to x:

$$T(x'|x) = T(x|x'), \tag{2.25}$$

(the $T(x'|x)$ is read "likelihood of proposing x' *given* x, i.e. given x is the current state).

2.8 Proof of Metropolis Monte Carlo

The Metropolis Monte Carlo method acts by generating a Markov chain. Markov chains were introduced in the preceding chapter and can be recapped as being sequences of states where the probability of moving from one state to the next state depends only on the current state, not on the earlier history of the chain.

To prove that the Metropolis Monte Carlo method generates a Boltzmann distribution, we start by considering the detailed balance condition. The detailed balance condition states that for any two states x and x' in the state space, the probability of being in state x and moving to state x' must be equal to the probability of being in state x' and moving to state x:

$$P(x)T(x'|x)A(x'|x) = P(x')T(x|x')A(x|x'), \tag{2.26}$$

where $A(x'|x)$ is the acceptance probability of moving from x to x'.

Substituting the Metropolis acceptance probability:

$$A(x'|x) = \min\left(1, \frac{P(x')}{P(x)}\right) \tag{2.27}$$

into the detailed balance condition, we consider two cases:

Case 1

If $\frac{P(x')}{P(x)} \leq 1$, then $A(x'|x) = \frac{P(x')}{P(x)}$, and the detailed balance condition becomes:

$$P(x)T(x'|x)\frac{P(x')}{P(x)} = P(x')T(x|x'), \tag{2.28}$$

which simplifies to:

$$T(x'|x)P(x') = P(x')T(x|x'), \tag{2.29}$$

which is true if T is symmetric, confirming that the detailed balance condition is satisfied.

Case 2

If $\frac{P(x')}{P(x)} > 1$, then $A(x'|x) = 1$ and $A(x|x') = \frac{P(x)}{P(x')}$. The detailed balance condition becomes:

$$P(x)T(x'|x) = P(x')T(x|x')\frac{P(x)}{P(x')}, \tag{2.30}$$

which simplifies to:

$$P(x)T(x'|x) = T(x|x')P(x), \tag{2.31}$$

which is true if T is symmetric, and hence again confirming that the detailed balance condition is satisfied.

Detailed balance ensures that $P(x)$ is a stationary distribution of the Markov chain. It does not, by itself, guarantee convergence to this distribution. For convergence from an arbitrary initial state, the chain must additionally be ergodic. In practice, this requires that the proposal mechanism allow all relevant regions of state space to be reached and that the chain not be confined to a periodic cycle. Under these conditions, $P(x)$ is the unique stationary distribution, and the Markov chain converges to $P(x)$ independently of its initial state.

Now, is there a catch here? Did we have to know $P(x)$ to evaluate the acceptance probability, $A(x'|x)$, in order to have the Markov chain ultimately sample $P(x)$? Well, not quite. If you inspect, $A(x'|x)$:

$$A(x'|x) = \min\left(1, \frac{P(x')}{P(x)}\right) \tag{2.32}$$

you will notice that it is only the ratio between $P(x')$ and $P(x)$ which must be known. Inserting the expression, Eq. (2.11), we have:

$$\begin{aligned} A(x'|x) &= \min\left(1, \frac{\frac{1}{Q}\exp\left(-\frac{V(x')}{k_BT}\right)}{\frac{1}{Q}\exp\left(-\frac{V(x)}{k_BT}\right)}\right) \\ &= \min\left(1, \exp\left(-\frac{V(x')-V(x)}{k_BT}\right)\right) \end{aligned} \tag{2.33}$$

and thus need only evaluate the energy change, $V(x') - V(x)$, going from state x to x', which can be done without knowing $P(x)$ entirely.

2.9 Metropolis Monte Carlo in practice

When implementing the Metropolis Monte Carlo method, the following procedure is commonly used:

1. Choose an initial state of the system.

2. Propagate the Markov chain for an initial equilibration period.

3. Discard the states generated during the equilibration period, since they may retain a dependence on the chosen initial state.

4. Continue propagating the Markov chain and estimate thermal averages using the subsequently generated states.

If a proposed move is rejected, the system remains in its current state. The current state must then be recorded again as the next state of the Markov chain. Failure to include these repeated occurrences would, in general, bias estimates of thermal averages.

Successive states of a Markov chain are generally correlated. As a result, the statistical uncertainty of a Monte Carlo estimate is usually larger than it would be for the same number of independent samples. These correlations should therefore be taken into account when estimating statistical uncertainties.

2.10 Importance sampling and reweighting

Sometimes it is convenient to construct a sample for one system with potential $V_0(x)$, while you are actually interested in investigating another system with potential energy $V(x)$. The potential of the system of interest may be written:

$$V(x) = V_0(x) + \Delta V(x) \tag{2.34}$$

The corresponding Boltzmann factor is therefore

$$\exp\left(-\frac{V(x)}{k_B T}\right) = \exp\left(-\frac{V_0(x)}{k_B T}\right) \times \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \tag{2.35}$$

The configurational partition function of the system of interest is therefore

$$\begin{aligned} Q &= \int_{-\infty}^{\infty} \exp\left(-\frac{V(x)}{k_B T}\right) dx \\ &= \frac{\int_{-\infty}^{\infty} \exp\left(-\frac{V_0(x)}{k_B T}\right) dx}{\int_{-\infty}^{\infty} \exp\left(-\frac{V_0(x)}{k_B T}\right) dx} \int_{-\infty}^{\infty} \exp\left(-\frac{V(x)}{k_B T}\right) dx \\ &= \int_{-\infty}^{\infty} \exp\left(-\frac{V_0(x)}{k_B T}\right) dx \frac{\int_{-\infty}^{\infty} \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \exp\left(-\frac{V_0(x)}{k_B T}\right) dx}{\int_{-\infty}^{\infty} \exp\left(-\frac{V_0(x)}{k_B T}\right) dx} \\ &= Q_0 \left\langle \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right\rangle_0 , \end{aligned} \tag{2.36}$$

where Q_0 is the partition function for the reference system and $\langle \ldots \rangle_0$ denotes the thermal average taken with a Boltzmann sample for the reference system.

Now that we have the partition function for the system of interest written in terms of the partition function of another, known system, we may write the thermal average of an observable, $G(x)$ as:

$$\begin{aligned} \langle G(x) \rangle &= \frac{\int_{-\infty}^{\infty} G(x) \exp\left(-\frac{V(x)}{k_B T}\right) dx}{\int_{-\infty}^{\infty} \exp\left(-\frac{V(x)}{k_B T}\right) dx} \\ &= \frac{1}{Q} \times \int_{-\infty}^{\infty} G(x) \exp\left(-\frac{V(x)}{k_B T}\right) dx \\ &= \frac{1}{Q} \times \frac{Q_0}{Q_0} \times \int_{-\infty}^{\infty} G(x) \exp\left(-\frac{V_0(x) + \Delta V(x)}{k_B T}\right) dx \\ &= \frac{1}{Q} \times \frac{Q_0}{Q_0} \times \int_{-\infty}^{\infty} \left[G(x) \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right] \exp\left(-\frac{V_0(x)}{k_B T}\right) dx \\ &= \frac{Q_0}{Q} \times \frac{\int_{-\infty}^{\infty} \left[G(x) \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right] \exp\left(-\frac{V_0(x)}{k_B T}\right) dx}{\int_{-\infty}^{\infty} \exp\left(-\frac{V_0(x)}{k_B T}\right) dx} \\ &= \frac{Q_0}{Q} \left\langle G(x) \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right\rangle_0 \\ &= \frac{1}{\left\langle \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right\rangle_0} \left\langle G(x) \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \right\rangle_0 \end{aligned} \tag{2.37}$$

where the derived expression for the partition function, Eq. (2.36), has been inserted in the last line. We thus have a means to evaluate the thermal average of an observable for a system when only knowing the thermal probability distribution for another system.

Beware that if the overlap in the probability distributions of the known system (that having $V_0(x)$) is very different from that of the unknown system (the one with $V(x)$), the method might be inefficient.

This procedure is commonly called reweighting: configurations sampled from the reference system are assigned weights

$$w(x) = \exp\left(-\frac{\Delta V(x)}{k_B T}\right) \tag{2.38}$$

to estimate averages in the target system.